

# QSSI Interface Manual

## cm\_mcu\_hwtest Firmware — MDT-TP CM Prototype MCU

### 1. Overview

The QSSI (Quad Synchronous Serial Interface) provides access to the external SPI/QSPI flash memory device. It is implemented on top of the TM4C1294 SSI1 peripheral using TivaWare's advanced SSI mode, which supports both standard SPI (single-bit data) and Quad SPI (four-bit data) with hardware CS (FSS) control.

**Connected device:** IS25LP064A-JBLA3 — ISSI 64 Mbit SPI/QSPI NOR Flash

### 2. Hardware Configuration

| Signal | MCU Pin | GPIO      |
|--------|---------|-----------|
| CLK    | PB5     | SSI1CLK   |
| FSS/CS | PB4     | SSI1FSS   |
| XDAT0  | PE4     | SSI1XDAT0 |
| XDAT1  | PE5     | SSI1XDAT1 |
| XDAT2  | PD4     | SSI1XDAT2 |
| XDAT3  | PD5     | SSI1XDAT3 |

- **Protocol:** Motorola SPI Mode 0 (CPOL=0, CPHA=0)
- **Data width:** 8 bits
- **Default bit rate:** 15 MHz
- **Supported bit rate:** 2 kHz – 50 MHz
- **Timeout:** 500 loop iterations (~10 µs each)

### 3. Commands

#### 3.1 qssi-s — Set Up the QSSI Port

```
qssi-s PORT FREQ
```

| Parameter | Description                            |
|-----------|----------------------------------------|
| PORT      | QSSI port number. Only 1 is supported. |

| Parameter | Description                                   |
|-----------|-----------------------------------------------|
| FREQ      | Bit rate in Hz. Valid range: 2000 – 50000000. |

Example — set to 1 MHz:

```
qssi-s 1 1000000
```

Must be called after power-on before any `qssi` access. The default bit rate at startup is 15 MHz.

3.2 `qssi` — QSSI Access (Read / Write)

```
qssi PORT MODE RW END DATA...
```

| Parameter | Description                                                        | Values                                         |
|-----------|--------------------------------------------------------------------|------------------------------------------------|
| PORT      | QSSI port number                                                   | 1                                              |
| MODE      | SPI data mode                                                      | 0 = standard SPI (1-bit), 1 = Quad SPI (4-bit) |
| RW        | Transfer direction                                                 | 0 = write, 1 = read                            |
| END       | Release CS after                                                   | 0 = hold CS, 1 = release CS                    |
| DATA...   | Write: bytes to send (up to 28). Read: number of bytes to receive. |                                                |

Numbers can be given in decimal or hex (0x prefix).

4. CS (Chip Select) Control

The `END` flag controls the hardware FSS line:

- `END=1`: CS is released after the last byte. Use for self-contained single-phase transactions (e.g. Write Enable, Bulk Erase).
- `END=0`: CS is held low after the transfer. Use when the next `qssi` call must belong to the same transaction (e.g. sending a read command before clocking out data).

**Important:** When `END=0`, the driver does not wait for the SSI to become idle before returning. Issue the next command only after a short pause to avoid a timeout error on the following read.

5. IS25LP064A Flash Command Reference

## 5.1 Identification

**Read JEDEC ID** — verify the device is responding:

```
qssi 1 0 0 0 0x9F
qssi 1 0 1 1 3
```

Expected: 0x9D 0x60 0x17 (Manufacturer: ISSI, Type: SPI Flash, Capacity: 64 Mbit)

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## 5.2 Status Register

**Read Status Register** (opcode 0x05):

```
qssi 1 0 0 0 0x05
qssi 1 0 1 1 1
```

| Bit | Name | Description                                        |
|-----|------|----------------------------------------------------|
| 0   | WIP  | Write In Progress. 1 = busy (erase/write ongoing). |
| 1   | WEL  | Write Enable Latch. 1 = write enable active.       |

Poll until WIP = 0 after any erase or write operation before issuing the next command.

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## 5.3 Write Enable

Must be issued before every erase and every page program. The WEL bit is automatically cleared by the flash after the operation completes.

```
qssi 1 0 0 1 0x06
```

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## 5.4 Erase Operations

Flash bits can only transition from 1 to 0 during a write. To restore bits to 1, a sector or bulk erase is required first.

**Sector Erase (4 KB)** — opcode 0x20, 3-byte address:

```
qssi 1 0 0 1 0x06
qssi 1 0 0 1 0x20 0xAA 0xBB 0xCC
```

Poll WIP until clear (~80 ms typical).

**Bulk Erase (full chip)** — opcode `0x60`:

```
qssi 1 0 0 1 0x06
qssi 1 0 0 1 0x60
```

Poll WIP until clear (~30 s typical).

## 5.5 Read Operations

**Standard Read** (opcode `0x03`, up to 33 MHz):

```
qssi 1 0 0 0 0x03 0xAA 0xBB 0xCC
qssi 1 0 1 1 N
```

**Fast Read** (opcode `0x0B`, up to 108 MHz, requires 1 dummy byte after address):

```
qssi 1 0 0 0 0x0B 0xAA 0xBB 0xCC 0x00
qssi 1 0 1 1 N
```

**Quad Output Fast Read** (opcode `0x6B`, data phase in quad mode):

```
qssi 1 0 0 0 0x6B 0xAA 0xBB 0xCC 0x00
qssi 1 1 1 1 N
```

## 5.6 Write Operations (Page Program)

A page is 256 bytes. Writes cannot cross a page boundary — if address + length exceeds the boundary, the write wraps within the same page. The target sector must be erased before writing.

**Standard Page Program** (opcode `0x02`):

```
qssi 1 0 0 1 0x06
qssi 1 0 0 1 0x02 0xAA 0xBB 0xCC 0xD1 0xD2 0xD3
```

Poll WIP until clear (~0.8 ms typical).

**Quad Page Program** (opcode `0x32`, data phase in quad mode):

```
qssi 1 0 0 1 0x06
qssi 1 0 0 0 0x32 0xAA 0xBB 0xCC
qssi 1 1 0 1 0xD1 0xD2 0xD3 0xD4
```

Poll WIP until clear.

## 6. Typical Session

```
# 1. Set bit rate to 10 MHz
qssi-s 1 10000000

# 2. Check device ID
qssi 1 0 0 0 0x9F
qssi 1 0 1 1 3
# Expected: OK. Data: 0x9d 0x60 0x17

# 3. Erase sector at address 0x000000
qssi 1 0 0 1 0x06
qssi 1 0 0 1 0x20 0x00 0x00 0x00
# Poll WIP until 0:
qssi 1 0 0 0 0x05
qssi 1 0 1 1 1

# 4. Write 4 bytes to address 0x000000
qssi 1 0 0 1 0x06
qssi 1 0 0 1 0x02 0x00 0x00 0x00 0xDE 0xAD 0xBE 0xEF
# Poll WIP until 0:
qssi 1 0 0 0 0x05
qssi 1 0 1 1 1

# 5. Read back 4 bytes from address 0x000000
qssi 1 0 0 0 0x03 0x00 0x00 0x00
qssi 1 0 1 1 4
# Expected: OK. Data: 0xde 0xad 0xbe 0xef
```

## 7. Source Files

| File                           | Description                              |
|--------------------------------|------------------------------------------|
| Firmware/Common/hw/qssi/qssi.c | Low-level SSI driver (init, read, write) |
| Firmware/Common/hw/qssi/qssi.h | tQSSI struct definition and prototypes   |

| File                                                 | Description                                   |
|------------------------------------------------------|-----------------------------------------------|
| Firmware/Projects/cm_mcu_hwtest/cm_mcu_hwtest_qssi.c | Command parser for qssi and qssi-s            |
| Firmware/Projects/cm_mcu_hwtest/cm_mcu_hwtest_io.c   | Hardware configuration (g_sQssi1 struct)      |
| Firmware/Projects/cm_mcu_hwtest/cm_mcu_hwtest.h      | Frequency limit constants (QSSI_FREQ_MIN/MAX) |